

Maximum Manhattan Distances

Time limit: 1.00 s

Memory limit: 512 MB

A set is initially empty and n points are added to it. Calculate the maximum Manhattan distance of two points after each addition.

Input

The first line has an integer n : the number of points.

The following n lines describe the points. Each line has two integers x and y . You can assume that each point is distinct.

Output

After each addition, print the maximum distance.

Constraints

- $1 \leq n \leq 2 \cdot 10^5$
- $-10^9 \leq x, y \leq 10^9$

Example

Input	Output
5 1 1 3 2 2 4 2 1 4 5	0 3 4 4 7